

My initial post focused on comparing the paradigms of Industry 4.0 and Industry 5.0. In her response, Sarra highlighted a common critique of Industry 5.0: that it is merely an extension of Industry 4.0 rather than a genuinely new paradigm. After further research, I found this argument to be partially valid. Industry 5.0 emerged shortly after Industry 4.0, and its relevance became even more evident during the COVID-19 pandemic (Javaid *et al.*, 2020). Consequently, Industry 5.0 can indeed be viewed as an evolution of Industry 4.0, incorporating social and environmental considerations (European Commission. Directorate General for Research and Innovation., 2020).

On the other hand, the goals of Industry 4.0 are structurally aligned with the optimisation of business models, often overlooking the climate crisis and failing to address deeper social tensions. In this sense, the Industry 4.0 paradigm may itself contribute to some of today's societal challenges, making it insufficient to meet contemporary societal needs (European Commission. Directorate General for Research and Innovation., 2021). Therefore, Industry 5.0 should not be understood primarily as a technological leap forward. Rather, its significance lies in redirecting industry towards broader societal objectives, including “a net-zero emissions economy, inclusive growth, health, social and regional cohesion, sustainable mobility and environmental regeneration” (European Commission. Directorate General for Research and Innovation., 2021: 6).

In Ross' discussion of Industry 5.0 in the software and digital infrastructure sector, he highlighted the importance of human-centricity and data quality for effective AI implementation. This prompted further reflection on how Industry 5.0 principles could improve data quality in smart factories. As Wang, Wu and Wang (2025) argued, data quality is a systemic issue involving systems, people and knowledge, making Industry 5.0's human-centred and sustainable approach highly relevant to reliable data-driven environments.

Similarly, Ariel emphasised how AI-based quality control systems can improve production line reliability. However, achieving greater reliability requires more than automation alone. Quality 5.0 therefore extends beyond the automation-driven Quality 4.0 approach by emphasising employee expertise, collaboration and adaptability, with workers acting not only as operators but also as supervisors capable of responding to unexpected disruptions (Olaitan and Elshennawy, 2024).

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